

TFLite

On-device sign inference for ESP32-P4 with TensorFlow Lite Micro. Captures frames from an IMX219 or OV5647 camera, runs the B-G colour-difference pipeline, classifies the sign, and sends the result to the ESP32-S3 over UART.

Camera Selection

Edit `image_provider.h`:

```
#define CAMERA_TYPE CAMERA_TYPE_IMX219 // or CAMERA_TYPE_OV5647 /  
CAMERA_TYPE_AUTO
```

- **IMX219**: MIPI CSI, 1536×1232 RAW10. Library builds at `IMG_SIZE` (96×96 default) via the override in `image_provider.h`.
- **OV5647**: MIPI CSI, 1920×1080 RAW10. The library stays at its own 160×160 default — `GetImage()` resizes to `IMG_SIZE` at runtime.
- **AUTO**: probes the shared I2C bus at startup (IMX219=0x10, OV5647=0x36) and picks whichever responds — no recompile needed when swapping cameras.

Both cameras feed the same pipeline, so switching cameras only requires changing `CAMERA_TYPE`.

Preprocessing Mode

Edit `image_provider.h`:

```
#define PREPROCESS_MODE PREPROCESS_MODE_BG // or GRAY / RGB
```

Mode	What <code>GetImage</code> produces	Use case
<code>PREPROCESS_MODE_BG</code> (default)	B-G colour difference + blob auto-crop + contrast	Sign inference (matches <code>AItraining/image_preprocess.py</code>)
<code>PREPROCESS_MODE_GRAY</code>	Plain BT.601 grayscale + contrast, no colour difference	Inference on models trained with plain grayscale
<code>PREPROCESS_MODE_RGB</code>	No inference — streams WB-corrected <code>IMG_SIZE×IMG_SIZE×3</code> RGB to Serial (<code>0xAA 0x55</code> <code>0xAA</code> + payload)	Data collection for Altraining, like the <code>IMX219_RGB_Serial</code> example

Pipeline

Camera → dynamic Gray-World AWB → B-G extraction → blur → contrast stretch → blob detection → auto-crop → resize IMG_SIZExIMG_SIZE → contrast → TFLite int8 inference → UART to S3.

See the repo-root `CLAUDE.md` for the full 13-step pipeline. Any change to `image_provider.cpp` must be mirrored in `AItraining/image_preprocess.py`.

UART Protocol (P4 ↔ S3, bidirectional)

Both directions share one UART at 921600 baud (P4 pins: RX=10, TX=11).

P4 → S3 — inference packet, 9 bytes:

Offset	Field
0-1	Sync <code>0xAA 0x55</code>
2	msg_type <code>0x01</code>
3-4	frame_id (uint16 LE)
5	label_id
6	confidence
7	flags (2 = sign_ready)
8	checksum (XOR of bytes 0-7)

S3 → P4 — control packet, 5 bytes:

Offset	Field
0-1	Sync <code>0xAA 0x55</code>
2	msg_type <code>0x02</code>
3	command
4	checksum (XOR of bytes 0-3)

Commands:

- `0x01 ACK_STOP` — S3 confirmed the sign → P4 stops transmitting (inference continues)
- `0x02 RESUME` — S3 done → P4 resumes transmitting

TX gating is a `volatile bool s_transmit_enabled`; when disabled the TX task drains its queue without writing. GPIO10 RX noise is mitigated with `INPUT_PULLDOWN`.

FreeRTOS Tasks

Task	Core	Priority	Stack	Role
<code>tflm</code>	1	3	32 KB	Capture → pipeline → inference → queue packet

Task	Core	Priority	Stack	Role
uart_tx	0	2	4 KB	Send packets; drops when <code>s_transmit_enabled == false</code>
uart_rx	0	2	4 KB	Read S3 control packets; toggle <code>s_transmit_enabled</code>
sd	0	1	8 KB	Write PGM frames to SD

Configurable Parameters

Param	Default	Description
CAMERA_TYPE	IMX219	CAMERA_TYPE_IMX219 / _OV5647 / _AUTO (I2C probe)
PREPROCESS_MODE	BG	PREPROCESS_MODE_BG / _GRAY / _RGB
kDebugBaud	921600	Debug serial baud
kUartBaud	921600	P4↔S3 UART baud
kEnableSdLogger	true	Save frames to SD
kSaveEveryNFrames	10	Save every N frames